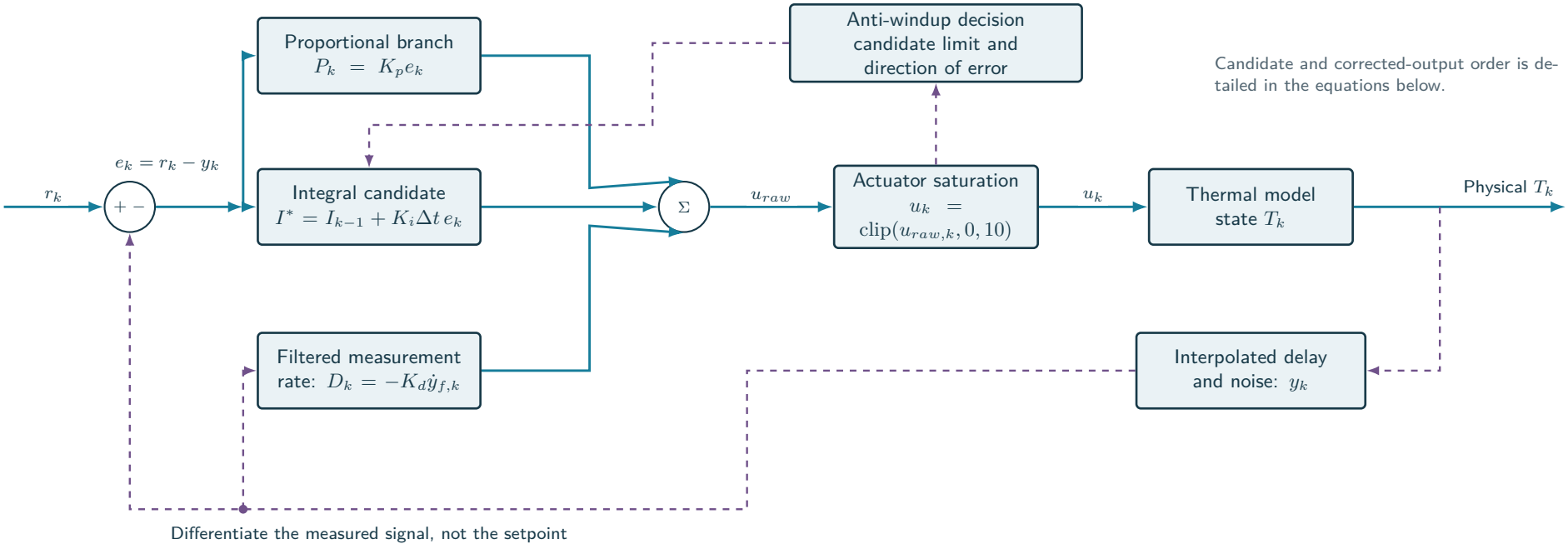
PID implementation: gain convention, filtering and saturation

Independent gains: heater/°C, heater/(°C s), heater s/°C. Derivative filter: $\dot{y}_{f,k} = a\dot{y}_{f,k-1} + (1-a)(y_k - y_{k-1})/\Delta t$, with $a = \tau_f/(\tau_f + \Delta t)$.



IMC source-derived tuning

$K_p = \dfrac{\tau + \theta/2}{K(\lambda + \theta)}; \quad T_i = \tau + \theta/2.$

$T_d = \dfrac{\tau\theta}{2\tau + \theta}.$

Parallel gains: $K_i = K_p/T_i$, $K_d = K_p T_d$. Time constants and gains must not be interchanged.

Conditional integration

Compute $u^* = P_k + I^* + D_k$. Retain I_{k-1} if $u^* > u_{max}$ with $e_k > 0$, or $u^* < u_{min}$ with $e_k < 0$. Otherwise accept I^*. Then recompute and clip the command.

Back-calculation option

$I_k = I^* + \dfrac{\Delta t}{T_t}\left[\mathrm{clip}(u^*) - u^*\right].$

Recompute $P_k + I_k + D_k$, then clip. The implementation requires $T_t \geq \Delta t$. Its numerical choice is an assumption.

Control semantics checked against FixedPID._calculate and imc_pid_gains. Fresh controller objects are required for independent runs because integral, derivative and measurement histories persist. No hardware protection is modeled.

Original explanatory drawing for ThermoFuzzy-PID. Source traceability: Cordeiro (UFC, 2021), documented repository equations and explicit reconstruction choices.